Project Title

Spatial and Statistical Analyses of the Mortality Rates of COVID-19 in 2020 in Illinois, USA

Introduction

This project investigates spatial patterns of COVID-19 mortality in Illinois using vector-based GIS analysis. Through the integration of public health data and geospatial tools in ArcGIS Pro, the project applies spatial statistics, bivariate classification, and regression techniques to reveal geographic disparities and the potential influence of poverty and social vulnerability. The analysis is divided into four core tasks: data preparation, spatial data management, spatial analysis, and statistical modeling.

Data Sets Used

The following datasets were used:

COVID-19 Mortality (2020) – Downloaded from the Illinois Department of Public Health as a PDF and manually converted into CSV format. Other datasets used included 2020 population data for Illinois counties, Social Vulnerability Index (SVI) and Percent in Poverty, County and State boundary shapefiles. All data was managed in a geodatabase and prepared for spatial analysis using ArcGIS Pro.

Methodology

The COVID-19 death data was extracted from a multi-cause mortality PDF using Notepad and Excel. Only COVID-19 rows were kept. The 2020 population values were used to calculate Mortality Rate per 100,000 using: (Mortality / Population) * 100,000.

The cleaned and structured dataset was saved as a CSV file for joining with the GIS layers.

Both state and county shapefiles were projected to NAD 1983 State Plane Illinois West FIPS 1202 (US Feet) to ensure consistency. A new file geodatabase (Illinois.gdb) was created, and the projected state and county layers were imported into it as feature classes.

Spatial clustering of COVID-19 mortality was assessed using two main tools:

- Global Moran's I was run four times using different spatial conceptualizations: Inverse Distance, Inverse Distance Squared, Contiguity (Edges Only), and Contiguity (Edges and Corners), as displayed in table 1. These tests evaluated whether high or low mortality values were spatially clustered or dispersed.
- Local Moran's I (Cluster and Outlier Analysis) was used to detect specific counties that formed significant clusters (e.g., High-High, Low-Low) or outliers (e.g., High-Low, Low-High). Only statistically significant results ($p \leq 0.05$) were reported in Table 2 and mapped in Figure 1. The final map included symbology for cluster types and was exported with a white background and no basemap.

Table 1. Spatial statistics from the Spatial Autocorrelation Analysis using Global Moran's I geoprocessing tool.

Conceptualization of Spatial Relationships	**Global Moran's I**	**Z-score**	**p-value**	**Distance Threshold (feet)**
Inverse Distance	0.143567	1.774401	0.075997	141291.7012
Inverse Distance Squared	0.145578	1.768281	0.077014	141291.7012
Contiguity Edges only	0.086835	1.549110	0.121355	
Contiguity Edges Corners	0.085163	1.555826	0.119749	

Table 2. Spatial statistics from Cluster and Outlier Analysis with Contiguity Edges Corners

Feature ID	Local Moran's I	Z-Score	p-value	COType
4	0.646954	2.034296	0.02	LL
7	-0.776249	-2.059997	0.022	LH
8	0.756198	1.976373	0.026	HH
16	-0.125741	-2.104186	0.01	HL
19	0.852495	1.72131	0.036	LL
21	-0.074196	-1.658997	0.046	HL
22	0.784095	1.673698	0.042	LL
27	-0.526414	-1.654855	0.038	HL
30	0.486384	2.216858	0.024	HH
33	-0.085161	-1.874813	0.038	LH
37	-0.266728	-3.014023	0.002	LH
45	0.876439	2.520774	0.008	LL
47	1.307102	2.226639	0.008	LL
50	-0.338975	-2.289668	0.016	HL
56	0.915623	1.783234	0.034	LL
65	-0.62834	-1.633491	0.048	LH
66	-0.743525	-2.489319	0.008	LH
74	0.972719	1.623855	0.044	LL
86	-1.370766	-1.980592	0.032	LH
93	-0.136533	-1.79196	0.04	LH
99	0.733432	2.024207	0.022	LL
102	0.285425	1.983508	0.02	LL

Each variable was classified using Natural Breaks (Jenks) into 3 categories: Low, Medium, High. These were combined into 9 bivariate classes (e.g., High-High, Med-Low, etc.).

Each used a 3x3 color grid legend and state/county outlines (1.0 pt for counties, 2.0 pt for state boundary).

Results

Spatial Clustering: Global Moran's I showed low to moderate positive spatial autocorrelation depending on the conceptualization. Local Moran's I identified both hotspots (High-High) and cold spots (Low-Low). The Northern portion of the state is spatially clustered. Most of the counties have low mortality rate of COVID-19, forming a Low-to-Low (LL) cluster except Carroll which is HH cluster, Cook, Ford and LaSalle that are HL cluster, and Henry and Mercer which are also LH cluster. Several statistically significant outliers were found as well.

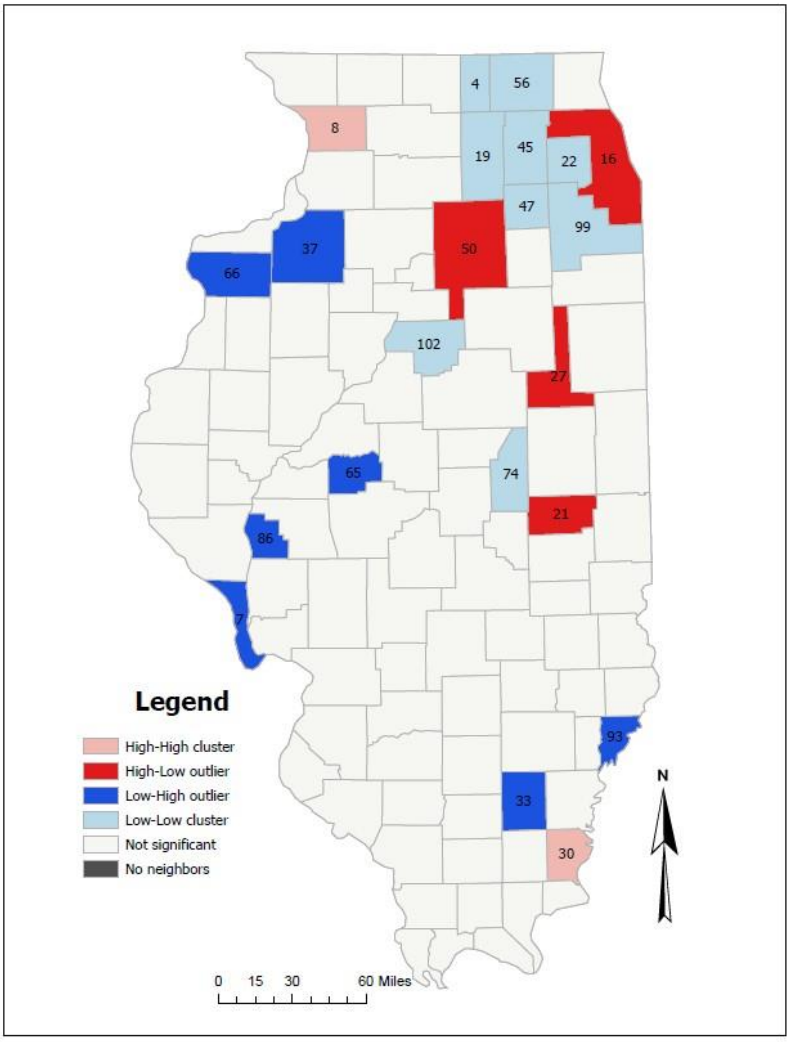

Figure 1. Cluster Map of COVID-19 Mortality

Bivariate Patterns: The Mortality vs. Poverty map showed more concentrated patterns of high mortality linked with high poverty. The Mortality vs. SVI map showed more dispersed patterns and less visual correlation.

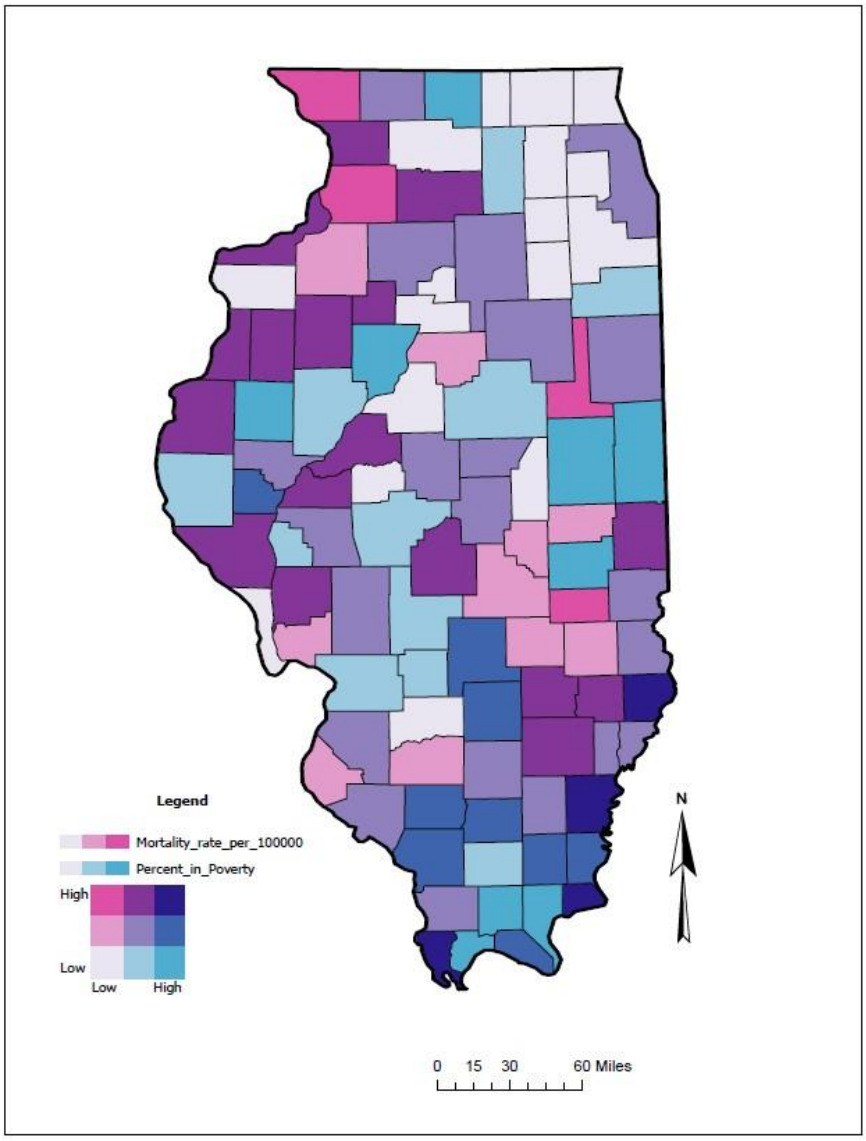

Figure 2. Mortality & Percent in Poverty

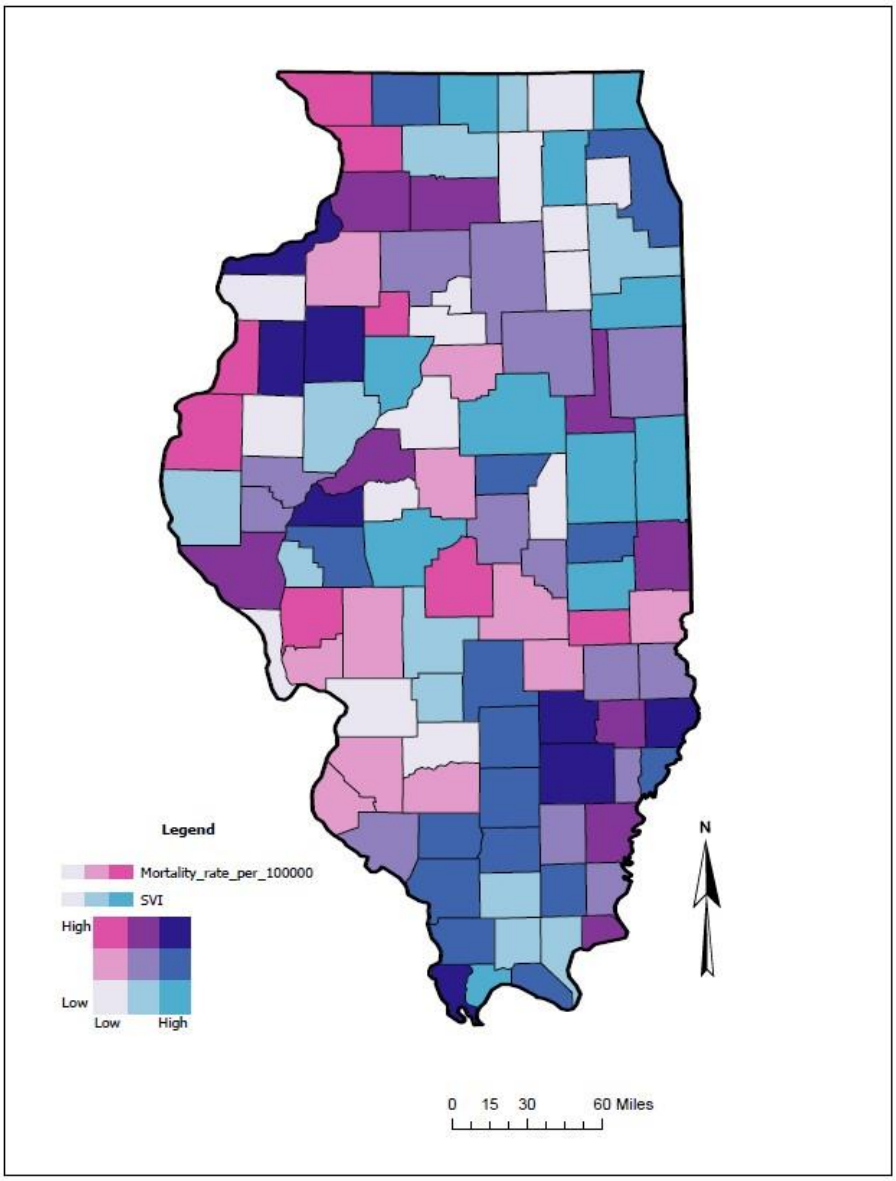

Figure 3. Mortality vs SVI

Linear Regression Results

Model	R	R^2	F	p-value

Mortality vs. Poverty	0.2493	0.0622	6.6297	0.0115
Mortality vs. SVI	0.1145	0.0131	1.3286	0.2518

Conclusion

This GIS-based analysis demonstrated that COVID-19 mortality in Illinois is not randomly distributed. Some counties experience significantly higher rates, forming spatial clusters. A modest positive correlation exists between poverty and mortality, while SVI showed little statistical effect. The use of bivariate mapping helped visualize these geographic relationships more clearly. Overall, the project successfully applied spatial statistics, classification, regression, and cartographic design to analyze a public health problem.